

Writing Small Numbers in Scientific Notation

MA.8.NSO.1.4

I can write very small numbers in scientific notation, read them back, and find how many times larger or smaller one number is than another.

Yesterday the exponents on the 10 were positive.

Today they are negative — and the numbers get very small.

Warm-Up

On your own. Four minutes. No notes.

1. Write 4,300,000 in scientific notation.
2. Write 10^{-3} as a fraction.
3. $\sqrt{95}$ is between which two whole numbers?
4. Write 0.007 as a fraction.

Warm-Up

Answers.

1. Write 4,300,000 in scientific notation. **4.3×10^6**
2. Write 10^{-3} as a fraction. **$\frac{1}{1,000}$**
3. $\sqrt{95}$ is between which two whole numbers? **9 and 10**
4. Write 0.007 as a fraction. **$\frac{7}{1,000}$**

Notes

Copy the table.

I. Negative powers of ten name the decimal places

10^0	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}
1	0.1	0.01	0.001	0.0001	0.00001

$$10^{-n} = \frac{1}{10^n} \qquad 10^{-3} = \frac{1}{1,000} = 0.001$$

A negative exponent puts the 1 in that decimal place: 10^{-4} is a 1 in the fourth place.

Notes

II. Small numbers in scientific notation

$$0.000007 = 7 \times 10^{-6} \qquad 0.0000672 = 6.72 \times 10^{-5}$$

$$3.4 \times 10^{-9} = 0.0000000034$$

The coefficient is still at least 1 and less than 10. The exponent is negative for a number less than 1.

The exponent is how many places the point moves: $0.0000672 \rightarrow 6.72$ is five places right, so 10^{-5} .

Notes

III. How many times larger — or smaller?

$$\frac{7.2 \times 10^{-4}}{1.8 \times 10^{-7}} = 4 \times 10^3 = 4,000$$

$$\frac{3 \times 10^8}{3 \times 10^{-4}} = 10^{12}$$

Divide the coefficients. Divide the powers of 10 — subtract the exponents. Multiply the results.

Which is bigger, 10^{-5} or 10^{-8} ? The one with the larger exponent: $-5 > -8$, so 10^{-5} .

Example 1

A quantity in words, to scientific notation.

The density of oxygen is 1,332 millionths of a gram per cubic centimeter.

Write this value as a decimal, then in scientific notation.

Example 1

Worked.

$$\frac{1,332}{1,000,000} = 0.001332 \quad \text{1,332 millionths, as a decimal}$$

$$0.001332 = 1.332 \times 10^{-3} \quad \text{the 1 is in the thousandths place}$$

Answer: $0.001332 = 1.332 \times 10^{-3}$

Your Turn

Same steps, your numbers. Boards up when done.

A human hair is about 67 hundred-thousandths of an inch thick.

Write this value in scientific notation.

Your Turn

Answer.

A human hair is about 67 hundred-thousandths of an inch thick.

Write this value in scientific notation.

0.00067 — the 6 is in the ten-thousandths place

Answer: 6.7×10^{-4}

Example 2

How many times larger?

How many times larger is 7.2×10^{-4} than 1.8×10^{-7} ?

Example 2

Worked.

$$\frac{7.2}{1.8} = 4 \quad \text{coefficients}$$

$$\frac{10^{-4}}{10^{-7}} = 10^{-4 - (-7)} = 10^3 \quad \text{powers of 10: } -4 - (-7) = 3$$

$$4 \times 10^3 = 4,000 \quad \text{multiply the two results}$$

Answer: 4,000 times larger

Your Turn

Same steps, your numbers. Boards up when done.

How many times larger is 5.2×10^{-3} than 2.6×10^{-8} ?

Your Turn

Answer.

How many times larger is 5.2×10^{-3} than 2.6×10^{-8} ?

2×10^5

Answer: 200,000 times larger

Whiteboards • Question 1 of 9

Boards up on three.

$$10^{-4}$$

Answer it.

Write it as a decimal.

Whiteboards • Question 1 of 9

Answer.

$$10^{-4}$$

a 1 in the fourth decimal place

Answer: 0.0001

Whiteboards • Question 2 of 9

Boards up on three.

Write 0.003 in scientific notation.

Answer it.

Which place is the 3 in?

Whiteboards · Question 2 of 9

Answer.

Write 0.003 in scientific notation.

3 thousandths

Answer: 3×10^{-3}

Whiteboards • Question 3 of 9

Boards up on three.

Write 2.5×10^{-4} in standard form.

Answer it.

Move the point four places left.

Whiteboards • Question 3 of 9

Answer.

Write 2.5×10^{-4} in standard form.

the 2 lands in the ten-thousandths place

Answer: 0.00025

Whiteboards • Question 4 of 9

Boards up on three.

A machine's cutting tolerance is 0.0000988 inch.

The part it cuts is 4 inches long.

Write the tolerance in scientific notation.

Answer it.

Every nonzero digit stays in the coefficient.

Whiteboards • Question 4 of 9

Answer.

A machine's cutting tolerance is 0.0000988 inch.

The part it cuts is 4 inches long.

Write the tolerance in scientific notation.

the 9 is in the hundred-thousandths place

Answer: 9.88×10^{-5}

Whiteboards • Question 5 of 9

Boards up on three.

Which is 0.00081 written in scientific notation?

A. 8.1×10^{-4}

B. 8.1×10^{-3}

C. 81×10^{-5}

D. 8.1×10^4

Answer it.

Whiteboards • Question 5 of 9

Answer.

Which is 0.00081 written in scientific notation?

A. 8.1×10^{-4}

B. 8.1×10^{-3}

C. 81×10^{-5}

D. 8.1×10^4

Answer: A — 8.1×10^{-4}

Whiteboards • Question 6 of 9

Boards up on three.

How many times larger is 6×10^{-3} than 2×10^{-5} ?

A. 300

B. 3

C. 30

D. 0.03

Answer it.

Whiteboards • Question 6 of 9

Answer.

How many times larger is 6×10^{-3} than 2×10^{-5} ?

A. 300

B. 3

C. 30

D. 0.03

Answer: A — 300

Whiteboards • Question 7 of 9

Boards up on three.

Which is greater, 3.9×10^{-5} or 7.4×10^{-8} ?

Answer it.

Compare the exponents first.

Whiteboards • Question 7 of 9

Answer.

Which is greater, 3.9×10^{-5} or 7.4×10^{-8} ?

$-5 > -8$, so 10^{-5} is the larger power

Answer: 3.9×10^{-5}

Whiteboards • Question 8 of 9

Boards up on three.

A computer displays a number as $4.2\text{E}-6$.

Write the number in standard form.

Answer it.

E-6 means $\times 10^{-6}$.

Whiteboards • Question 8 of 9

Answer.

A computer displays a number as $4.2\text{E}-6$.

Write the number in standard form.

$$4.2 \times 10^{-6}$$

Answer: 0.0000042

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Fishing line A is 2×10^{-2} inch thick.

Fishing line B is 6×10^{-2} inch thick.

Which is thicker, and how many times? Name a thickness between them.

Write your answer in sentences.

Same power of 10 — compare the coefficients.

Whiteboards • Question 9 of 9

Answer.

Fishing line A is 2×10^{-2} inch thick.

Fishing line B is 6×10^{-2} inch thick.

Which is thicker, and how many times? Name a thickness between them.

0.02 in and 0.06 in

Answer: B; 3 times; e.g. 4×10^{-2} in.

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Convert between standard and scientific notation: small numbers — 85M

Compare small numbers — KDR

Powers of ten with negative exponents — SKD